SERVICE-ORIENTED ARCHITECTURE (SOA) ENGINEERING PLANNING SUPPLEMENTAL GUIDE

Back to View

The materials presented in this technical guideline provide specific supplemental information into OUM for the purposes of SOA Engineering Planning. The SOA Engineering Planning Supplemental Guide contains the following components:

SOA Engineering Planning Overview

This component provides an overview of the concepts of SOA Engineering Planning.

SOA Engineering Planning Supplemental Task Guidelines

The Supplemental Task Guidelines should be used in conjunction with the standard OUM tasks in order to accomplish SOA Engineering Planning work at the detail task level.

SOA Engineering Planning Technique/Tool Guidelines

The Technique/Tool Guidelines provide Information on how to apply SOA Engineering Planning techniques/tools through the tasks in OUM.

PDF View

You may also prefer to access this information through the **SOA Engineering Planning Supplemental Guide (PDF View)** located in the Supplemental Guides section of the Supplemental Guidance page.

Back to Top

PURPOSE

The purpose of this document is to provide an overview of a SOA Engineering Planning engagement, and to provide implementation teams with *supplemental guidance* for the successful planning and execution of SOA Engineering Planning Engagements.

Back to Top

AUDIENCE

This document is intended for practitioners who will be planning SOA solutions whether or not the organization has chosen to use OUM for delivery. If the subsequent SOA project delivery is based on OUM, the work for this SOA Service Planning engagement can be simplified as OUM already incorporates the SOA delivery best practices in the SOA Project Delivery View.

Back to Top

HOW TO USE THIS DOCUMENT

This document contains supplemental guidance for planning and executing SOA Engineering Planning engagements and should be used in conjunction with and not instead of the OUM base method guidance. In many cases the work that needs to be done for the SOA Engineering planning process is fully contained within the OUM base method task and Technique files, therefore, no task supplement is needed for that task.

Back to Top

LIMITATIONS TO THIS DOCUMENT

This document does not describe OUM in detail – the reader must review the documentation and training materials referenced below to gain a basic understanding of each.

Back to Top

RELATED DOCUMENTATION AND TRAINING

The following related documentation & training provide additional detailed guidance related to OUM implementations and are prerequisites to this document:

- OUM Method Pack – Desktop Installable or Hosted version
- OUM Training and Accreditation Program

Back to Top

CONTRIBUTORS

Stephen Bennett, Jerome Bugnet, Sigrid Gylseth, Phil Head, Nicole Holthöfer, Jan Kettenis, Giustino Longo, Stephen Verba

Back to Top

ORACLE

Copyright © 2006, 2016, Oracle and/or its affiliates. All rights reserved.

SERVICE-ORIENTED ARCHITECTURE (SOA) ENGINEERING PLANNING OVERVIEW

Back to View

WHAT IS SOA ENGINEERING PLANNING?

The SOA Engineering Planning view and Supplemental Guide are used to help organizations adjust or complement their project delivery methodology to support SOA execution. Many enterprises struggle with SOA adoption. One area of particular concern is the delivery of projects within their SOA. Many enterprises attempt to apply traditional software engineering methodologies (such as Unified Process, agile, waterfall, etc) that have not been adopted to handle the wider, cross-project reuse that must be addressed for projects participating in a SOA strategy. The SOA Engineering Planning view supports a service where the organization is taken through a series of facilitated workshops aimed at developing their own enterprise methodology for SOA projects.

The SOA Engineering Planning view also looks at instrumentation for monitoring and management. Instrumentation is a fundamental of achieving success with a SOA strategy and spans a much broader area than simply instrumenting the runtime components of the SOA participants. Instrumentation for SOA includes collecting information in support of improving engineering efficiency and effectiveness, measuring and improving against SOA maturity level, understanding business value by linking this analysis to the underlying workings of SOA, in addition to assisting with making SOA operationally manageable. Therefore, this view looks into what kind of instrumentation is required to collect appropriate data through the new engineering processes.

As with all SOA planning views, this view is used to plan one particular aspect of the SOA strategy and can be combined with other SOA planning views based on the organization's needs.

It is important to note that OUM already has an approach for delivering SOA projects. The SOA Engineering Planning view should be used when the enterprise wants to develop their own enterprise-specific approach and adapt their own internal methodology to include such an approach. For more information on the OUM SOA approach, refer to the SOA Project Delivery view.

Background

Service-Oriented Architecture (SOA) is an Information Technology (IT) strategy that, in many cases, spans the entire enterprise. IT projects are implemented to fit within a service-oriented architecture and are therefore just a piece of the larger SOA initiative. Further, the extensive scope that a SOA may have across an enterprise requires that a repeatable process and sound engineering disciplines be applied through delivery cycles. This is especially true if external or offshore resources are to be utilized. Many enterprises struggle with inconsistency across deliveries for projects that need to coexist. This problem is also an inhibitor to reuse and agility and can be addressed by the adoption of a sound enterprise class SOA methodology. If projects are developed independently, without consistent tie-in and guidance from the overall SOA leadership team, the effectiveness and value of each delivery is substantially reduced. Fundamental to successful SOA adoption is the establishment of an enterprise methodology for SOA that is used to tie project delivery and operations into the overall SOA governance structure and reference architecture, as well as apply the best usage of investment in service infrastructure.

Back to Top

WHEN DO YOU USE THE SOA ENGINEERING PLANNING VIEW?

This view can best be used when the enterprise does not have an appropriate methodology for SOA. Typical indicators of the lack of an enterprise methodology for SOA are:

- cost and/or time explosion in projects delivering services
- service sprawl
- services with limited to no reuse
- inability of managing dependencies across service usage

It may also be used in combination with the SOA Modeling Planning view that is used when a modeling strategy is lacking. It is also recommended to be proposed as part of any strategic SOA engagement. It is an important part of the SOA governance effort and all organizations should at least consider it.

An engagement based on this view can be focused on the development of an enterprise methodology for SOA for the organization or may be combined with a pilot project delivery. If it is combined with project delivery, it is recommended to finalize the major areas of the engagement before starting the delivery of this phase within the project. This type of engagement is focused on creating a baseline for the organization. It is important that the organization takes over ownership to further evolve the methodology by reflecting feedback from projects.

Also, if the organization already has an appropriate methodology for SOA, but has not included any instrumentation for monitoring and management to measure the effectiveness of the methodology, then you may choose to use parts of the view, specifically, the instrumentation and metrics, for each of the areas included in the view. As stated earlier, OUM already has an approach for delivering SOA projects. The SOA Engineering Planning view should be used when the enterprise wants to develop their own enterprise-specific approach and adapt their own internal methodology to include such an approach. For more information on the OUM SOA approach, refer to the SOA Project Delivery view.

Back to Top

POSITIONING, QUALIFICATION AND SCOPE

Service-Oriented Architecture (SOA) requires planning that spans the enterprise or area of focus within the business that is applying it. Projects may not be simply architected, constructed and delivered independently in an effective SOA. Consistency and planning must be established across deliveries to ensure that the delivery maximizes the benefit of the SOA, as well as contributes to the overall SOA effectiveness.

Enterprise SOA requires that the method for delivering projects that will utilize SOA must be executed with a consistent strategy from inception through delivery and into steady state. This consistent strategy can be either planned for and developed in advance of any project, or developed at the same time as delivering a pilot project. The focus of an engagement based in this view is the development of a enterprise-specific SOA approach separate from the delivery of a project. This allows the approach to be developed more thoroughly. The strategy can then be refined with each additional delivery going forward.

It is clear that enterprise SOA engineering spans more than simply identifying and engineering services. Aside from influencing the software development lifecycle in many areas, this framework also impacts operations, and the overall enterprise architecture.

The delivery process then becomes a natural extension of the adopted SOA, rather than separate entities with unique goals. Further, the delivery process for SOA-based projects needs to be augmented with additional activities in order to ensure the following:

- Requirements are harvested with an eye towards service-orientation.
- Potential shared services are identified at the correct time in the delivery lifecycle.
- Existing shared services applicable to the delivery are discovered and applied in the application architecture correctly.
- Appropriate shared service testing strategies are adhered to.
- The deployment strategy is identified for shared services resulting from the delivery.
- New services are defined in accordance with the established rules of the governing SOA.
- Services are designed in accordance with the design policies established by the governing SOA, including:
    - service versioning strategy
    - implementation guidelines

    - coding standards
    - management enablement capabilities
- The application and the new resulting shared services are transitioned to operational state smoothly.
- The lifecycle of services is managed appropriately.
- SOA asset usage is tracked, measured and enforced.

Instrumentation, with respect to business and particularly IT has been around for a long time. However, it has not generally been utilized as a core component of any particular IT strategy. Quite often, it is introduced as an afterthought, or as a missed requirement. Many view the addition of instrumentation to be a bottleneck or unnecessary overhead. This reasoning typically stems from those who have been forced to add instrumentation as a reactive measure after a project has been completed.

When addressed up front and architected for accordingly such as we suggest in this view, instrumentation may be engineered to have as little footprint as possible, and in some cases is minimal. This is rarely the case when it is added in band-aid fashion after project completion.

When applied correctly, the cost of instrumentation related to the SOA engineering cycle pales in comparison to the benefits. From a business and IT perspective instrumentation provides the following (to name a few):

- Assistance with understanding IT performance and generated business value.
- Justification data for considered strategy changes.
- Information indicating areas for improvement resulting in greater engineering and operational efficiency.
- Assistance with critical internal and external conformance pressures.
- Measurement data leading to improved business process effectiveness.
- Data supporting IT strategy maturity.

When applied correctly, instrumentation provides the same benefits at a business level that we, as consumers of instrumentation use it for:

- **Conformance:** The ability to operate within predefined values.
- **Performance:** The ability to measure performance and gather the information necessary to improve.
- **Competitive Advantage:** The right information collected for the business allowing management to make better business, IT and operational decisions.

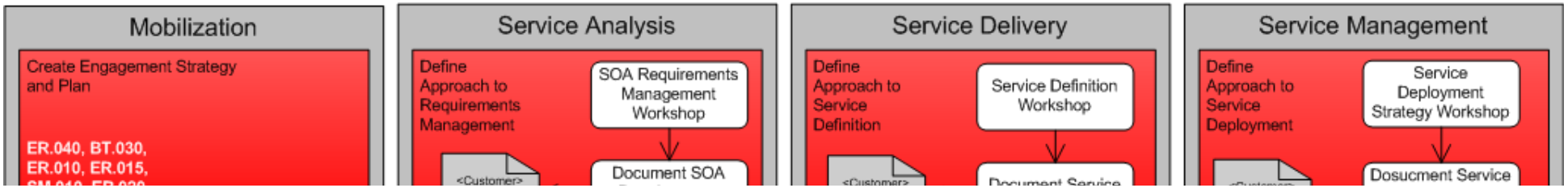
This SOA Engineering Planning view places instrumentation as a core component of the SOA strategy. It emphasizes instrumentation as a discipline required to more effectively implement an SOA strategy; and includes, but is not limited to just the projects and applications built within. By utilizing instrumentation as part of the SOA strategy, IT's ability to support the business will become more natural. Further, the business's understanding of the benefits provided by IT will become more apparent as IT becomes more proficient in responding to business demands. This is only possible if IT utilizes instrumentation to measure and collect performance data and then acts upon it to improve and mature as an organization.

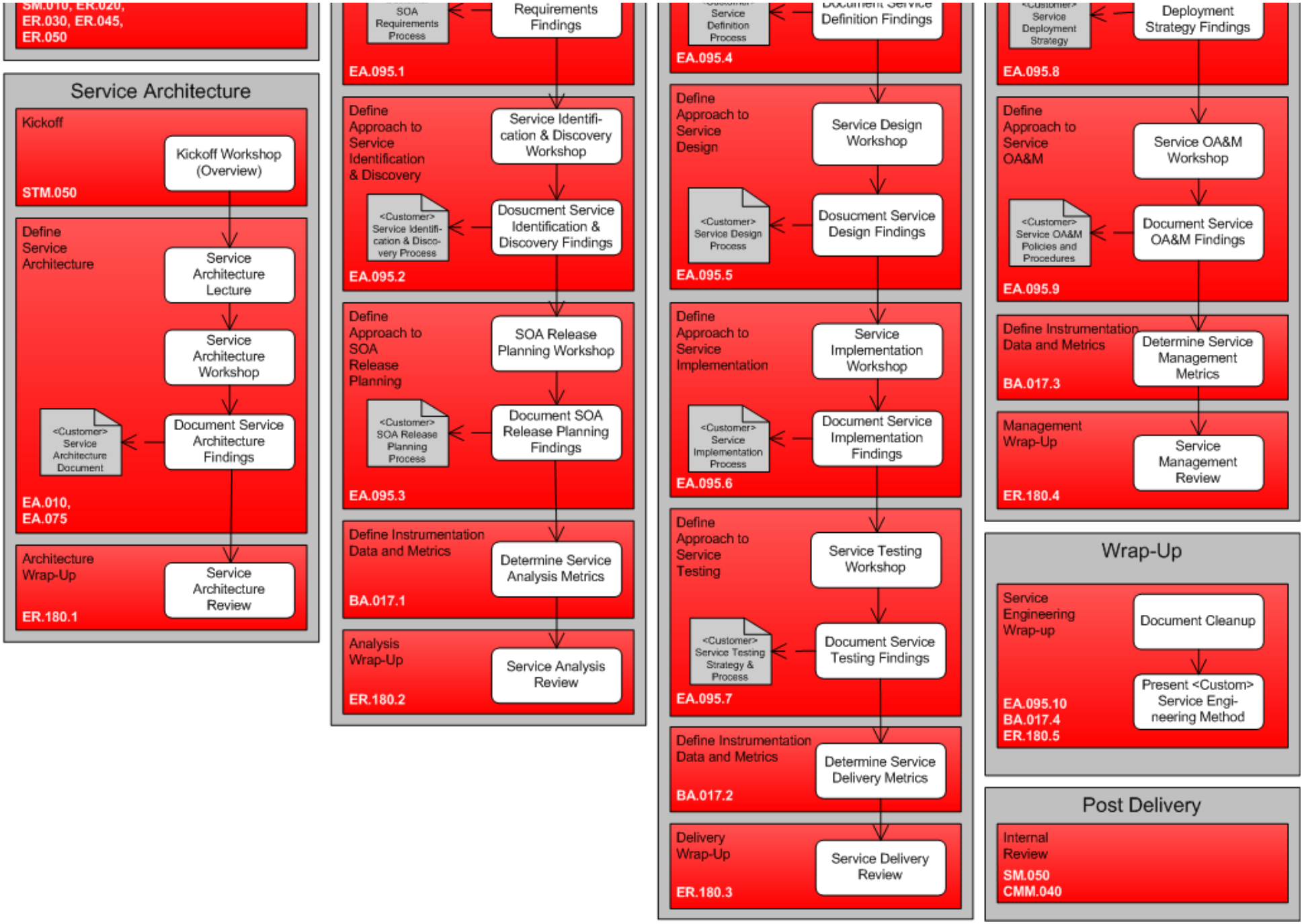
SOA, like any IT strategy, must be aligned to support the business needs. Most businesses have already established core business measurement metrics which are used to measure against defined business objectives. Due to increased IT spending over the last decade a common business objective is to reduce IT spending while still supporting the business strategy – IT is being asked to do more with less. IT, like other business units, must utilize metrics to measure its performance in support of the business.

Instrumentation for monitoring and management that is a part of this view represents an integral input into SOA strategy and planning with respect to defining the metrics which will be used to measure SOA execution and utilized to make improvements.

Back to Top

SOA ENGINEERING PLANNING OVERVIEW





There are seven areas in the SOA Engineering Planning view, each with its own set of tasks.

MOBILIZATION – Mobilization begins with the qualification of the effort to scope the engagement to best meet the needs of the organization. An engagement may be based solely on this view, but may also be increased to cover multiple views to produce a more full-featured planning and delivery engagement. In particular it may be combined with the SOA Modeling Planning view. It may also be scoped back to focus on only a subset of the six areas covered in this view, streams, tasks, and work products, while omitting others. Additional guidance on Mobilization can be found in the SOA Engineering Planning Task Guidelines.

SERVICE ARCHITECTURE – Service Architecture starts with the engagement kickoff and educating the organization in the concepts of a SOA engineering. It also helps familiarize them with the terminology that will be used throughout the engagement. Additional guidance on Service Architecture can be found in the SOA Engineering Planning Task Guidelines.

SERVICE ANALYSIS – During Service Analysis you develop the Service Analysis portion of the SOA engineering approach. The procedures and policies covered by Service Analysis coincide closely with the types of activities typical of traditional software engineering analysis practices but additionally take into account the needs of the SOA. When you have developed the Service Analysis portion of the SOA engineering approach, use it as an input to determine engineering metrics and instrumentation data related to Service Analysis that the organization feels is most important to them. Additional guidance on Service Analysis can be found in the SOA Engineering Planning Task Guidelines.

SERVICE DELIVERY – Service Delivery covers the core engineering processes and guidelines required to define, design, construct and test services. When you have developed the Service Delivery portion of the SOA engineering approach, use it as an input to determine engineering metrics and instrumentation data related to Service Delivery that the organization feels is most important to them. Additional guidance on Service Delivery can be found in the SOA Engineering Planning Task Guidelines.

SERVICE MANAGEMENT – Service Management covers the activities to transition and operate services. When you have developed the Service Management portion of the SOA engineering approach, use it as an input to determine engineering metrics and instrumentation data related to Service Management that the organization feels is most important to them. Additional guidance on Service Management can be found in the SOA Engineering Planning Task Guidelines.

WRAP-UP – During Wrap-Up, all work products are finalized for the engagement and prepared for project sign-off. The completed approach is presented back to the organization to wrap-up the engagement. Additional guidance on Wrap-Up can be found in the SOA Engineering Planning Task Guidelines.

POST DELIVERY - Following the successful completion of an engagement, a few post-delivery activities should take place, i.e., acceptance sign-off and documenting lessons learned. Additional guidance on Post Delivery can be found in the SOA Engineering Planning Task Guidelines.

Back to Top

WHAT OTHER VIEWS RELATE TO THE SOA ENGINEERING PLANNING VIEW?

Refer to the SOA in OUM page for access as well as descriptions for all the SOA views in OUM.

Back to Top

HOW DOES SOA ENGINEERING PLANNING RELATE TO OUM?

The SOA Engineering Planning process has been mapped to the Envision focus area of Oracle Unified Method to help show which tasks would typically be performed during each of the six areas from Mobilization to Wrap-Up.

See the mapping document and SOA Engineering Planning view for applicable OUM tasks.

Back to Top

ORACLE

Copyright © 2006, 2016, Oracle and/or its affiliates. All rights reserved.

SERVICE-ORIENTED ARCHITECTURE (SOA) ENGINEERING PLANNING SUPPLEMENTAL TASK GUIDELINES

Back to View

This document contains OUM supplemental task guidance for the execution of a service based on the SOA Engineering Planning view.

- The first column of the following table provides a complete list of all tasks in this document for which "supplemental" guidelines are provided as well as a quick link to the "supplemental" task guideline in this document.
- The second column provides a link to the OUM Task Overview.

The task guidelines in this document should be used to supplement the guidelines that appear in the OUM Task Overview.

SOA Engineering Planning - OUM Supplemental Task Guidelines (this document) collapse table | expand table	OUM Task Overview
Mobilization	
PJM.SM.010 Define and Confirm Scope	PJM.SM.010 Define and Confirm Scope
ER.020 Determine Envision Engagement Strategy	ER.020 Determine Envision Engagement Strategy
ER.030 Set and Agree on Plan for Envision Engagement	ER.030 Set and Agree on Plan for Envision Engagement
Service Architecture	
PJM.STM.050 Conduct Team Orientation	PJM.STM.050 Conduct Team Orientation
EA.010 Review Architecture Principles, Guidelines and Standards	EA.010 Review Architecture Principles, Guidelines and Standards
EA.075 Develop Technology Reference Architectures	EA.075 Develop Technology Reference Architectures
ER.180.1 Prepare for and Present Solution Recommendations	ER.180 Prepare for and Present Solution Recommendations
Service Analysis	

EA.095 (.1,.2,.3) Review Enterprise Software Development Process	EA.095 Review Enterprise Software Development Process
BA.017.1 Determine Metrics Collection and Reporting Strategy	BA.017 Determine Metrics Collection and Reporting Strategy
ER.180.2 Prepare for and Present Solution Recommendations	ER.180 Prepare for and Present Solution Recommendations
Service Delivery	
EA.095 (.4,.5,.6,.7) Review Enterprise Software Development Process	EA.095 Review Enterprise Software Development Process
BA.017.2 Determine Metrics Collection and Reporting Strategy	BA.017 Determine Metrics Collection and Reporting Strategy
ER.180.3 Prepare for and Present Solution Recommendations	ER.180 Prepare for and Present Solution Recommendations
Service Management	
EA.095 (.8,.9) Review Enterprise Software Development Process	EA.095 Review Enterprise Software Development Process
BA.017.3 Determine Metrics Collection and Reporting Strategy	BA.017 Determine Metrics Collection and Reporting Strategy
ER.180.4 Prepare for and Present Solution Recommendations	ER.180 Prepare for and Present Solution Recommendations
Wrap-Up	
EA.095.10 Review Enterprise Software Development Process	EA.095 Review Enterprise Software Development Process
BA.017.4 Determine Metrics Collection and Reporting Strategy	BA.017 Determine Metrics Collection and Reporting Strategy
ER.180.5 Prepare for and Present Solution Recommendations	ER.180 Prepare for and Present Solution Recommendations
Post Delivery	
PJM.CMM.040 Document Lessons Learned	PJM.CMM.040 Document Lessons Learned

Back to Top

MOBILIZATION GUIDANCE

Mobilization begins with the qualification of the effort to scope the engagement to best meet the needs of the organization. An engagement may be based solely on this view, but may also be increased to cover multiple views to produce a more full-featured planning and delivery engagement. In particular it may be combined with the SOA Modeling Planning view. It may also be scoped back to focus on only a subset of the six areas covered in this view, streams, tasks, and work products, while omitting others.

Create Engagement Strategy and Plan

Use a set of interviews and meetings with the organization to assess the maturity of the organization with respect to SOA and specifically, SOA engineering. Clarify the objectives and scope of the engagement. Identify the stakeholders from the organization needed to participate and create a detailed project plan. Agree on the scope and plan with the organization.

Key Responsibilities

This area involves the following roles:

Role	Responsibility

Implementing Organization	
Enterprise Architect	Assess and understand the maturity of the organization and suggest a scope for the engagement. Be familiar with the material from SOA Reference Architecture Planning and all OUM techniques available for this engagement.
Enterprise Repository Manager	Support the enterprise architect on assessment and identify the topics relevant for the organization. Needs to be a specialist on OUM with project experience in SOA engineering.
Customer (User)	
Customer Project Sponsor	Agree on the scope of the engagement.
Customer Project Manager	Identify the stakeholders needed in the engagement and ensure their availability and commitment
Enterprise Architect	Provide insight into the SOA strategy and approach of the organization. Responsible for enterprise SOA or enterprise architecture in general at the organization.
Methodologist	Provide insight into the approach to SOA and how this is mapped into the Software Development Lifecycle (SDLC) of the organization. Responsible for the SDLC of the organization.

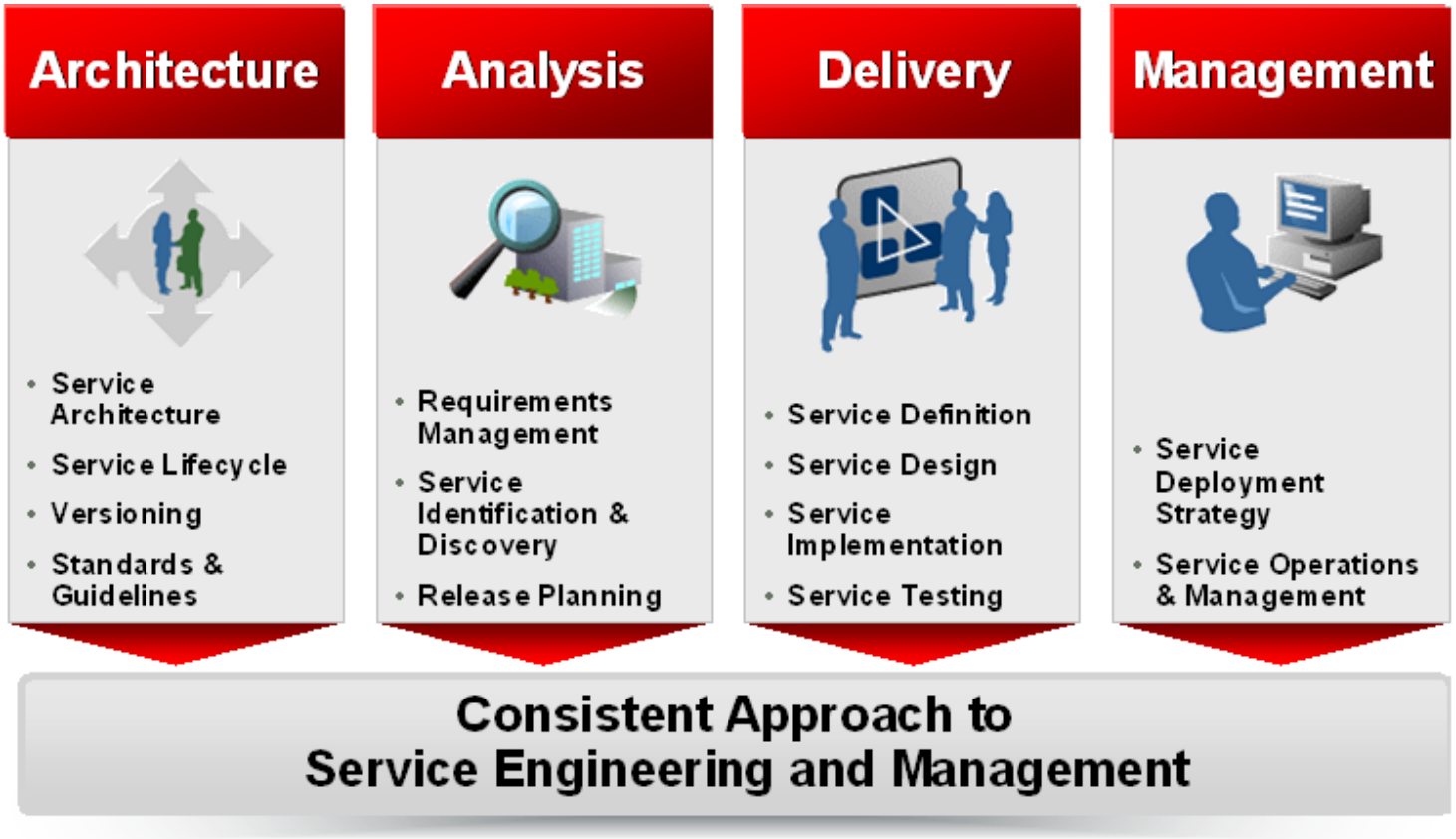
Back to Top

SM.010 Define and Confirm Scope

Determine Engagement Scope

The engagement lead is primarily responsible for determining the scope of the engagement. The areas and work products should be aligned with organization objectives, and the activities should support successful completion of the work products.

There are common areas of SOA engineering that are common across all SOA deployments and should be covered by any organization adopting SOA. The depiction below provides a summary of the OUM SOA engineering approach:



Topics covered by SOA engineering include:

- Service Architecture: Defines the basic terminology and taxonomy around services
- Service Lifecycle: Define the lifecycle of a services and the basic meta model needed to support collaboration across projects
- Versioning: Defines a common versioning strategy for services and identifies migration strategies to minimize impact on consumers
- Standards and Guidelines: Introduces an approach to identify and manage architectural policies for services. This approach can be used to standardize and harmonize the service landscape.
- Requirements Management: Provides a process for harvesting requirements in a manner that naturally facilitates service identification and discovery.
- Service Identification and Discovery: Establishes the procedures around identifying service candidates, as well as discovering reuse candidates from the existing service catalog. Takes the process from identification and discovery, through the justification processes required to determine if an existing service can be viable for reuse in the proposed manner, or if the proposed service candidate should be realized as a shared service.
- Release Planning: Provides the ground works necessary for planning for project and service deliveries within an SOA.
- Service Definition: Takes the identified service candidates and defines the service boundaries and resulting service contracts.
- Service Design: Provides the best practices and procedures required to design services from the service contract and produce the service interface.
- Service Implementation: Provides the guidelines for effectively and efficiently developing shared services.
- Service Testing: Defines the strategy that should be taken with respect to ensuring the appropriate level of quality for delivered shared services. The information obtained through service testing is also used to enable Service Deployment.
- Service Deployment: Defines the guidelines and practices that need to be considered when deploying services into a shared environment.
- Service Operation, Administration and Maintenance (OA&M): Covers the Operation, Administration and Maintenance guidelines required for supporting the SOA operational environment. OA&M is not simply about keeping the environment operational, but enabling reuse and evolution, in addition to measuring the SOA adoption and success.

With the proper engineering approach in place, the SOA can be started from the ground up and will grow effectively from project to project.

In addition to determining the topics for SOA engineering, you should strongly consider including instrumentation for monitoring and management related to the other subjects the organization chooses to include in the scope of the engagement. Instrumentation, with respect to business and particularly IT has been around for a long time. However, it has not generally been utilized as a core component of any particular IT strategy. Quite often it is introduced as an afterthought, or as a missed requirement. Many view the addition of instrumentation to be a bottleneck or unnecessary overhead. This reasoning typically stems from those who have been forced to add instrumentation as a reactive measure after a project has been completed.

When addressed up front and architected for accordingly such as we suggest in this view, instrumentation may be engineered to have as little footprint as possible, and in some cases is minimal. This is rarely the case when it is added in band-aid fashion after project completion. Therefore, it is recommended that you make strong arguments for including this as part of the scope of the engagement.

A number of meetings, phone calls, or emails can be used to finalize the scope with the organization. Organization objectives may originate from sources such as:

- Meetings
- Findings from previous SOA assessments
- Architect input from a current engagement

Back to Top

ER.020 Determine Envision Engagement Strategy
ER.030 Set and Agree on Plan for Envision Engagement

Determine Staffing, Schedule and Level of Effort

Applying SOA engineering to an enterprise typically takes four to eight weeks, possibly slightly longer when instrumentation is included in the engagement. Topic areas are developed through a series of facilitated workshops, where each topic is addressed with the appropriate leadership team members provided by the organization. In these workshops, the organization will need to demonstrate the current approach to that topic. Then OUM collateral is used to discuss best practices and methods. The organization approach will be refined based on the discussion. The policies and procedures developed through the workshops are then formalized through documentation of the findings. It will then be in the responsibility of the organization to apply needed changes to their documents.

Once the engagement has been scoped, the engagement lead should determine a level of effort (LOE) for each delivery area and stream, which will combine into an estimate for the engagement as a whole. A number of factors can affect the estimate in addition to the common factors, such as:

- the availability of enterprise materials on their SDLC or SOA engineering process documentation, etc.
- the availability and maturity of a SOA Reference Architecture
- the availability of existing experience regarding instrumentation for monitoring and management

If this material is not available, effort must be included to collect or even create this kind of material as it is an important input to this kind of engagement.

The following additional risk factors may have an impact on the estimation of effort:

- the overall maturity of project governance
- the maturity of the organization with respect to SOA
- the number of LOBs involved

Add additional effort if the organization needs to be introduced to basic SOA disciplines or if basic IT engineering methods need to be discussed.

Create and Deliver Exhibit

When determining the strategy for the engagement, you should consider the existing enterprise situation. The development of the strategy and a plan for the engagement

duration of the engagement can vary, depending on the size and complexity of the project. The strategy and plan should be concluded with proposal document or presentation to the organization where the go/no-go decision is received.

You can use the following questions as an assistance to scope the engagement, if not already answered through the previous steps:

- What are the primary goals for undertaking this engagement?
- What are the success criteria for this engagement?
- Are there specific dates that need to be met?
- How soon are you looking to get started?
- Roughly how many concurrent development projects does your IT organization run?
- How large and mature is your Enterprise Architect and project architect community?
- Has business strategy been defined, and if so, what method was used to define it (e.g., strategy map, balanced scorecard, etc.)?
- Have IT goals and initiatives been defined and mapped to business strategy?
- What SDLC processes or methodology do you follow?
- How important is SOA to the future of your IT organization?
- Has your enterprise architecture team attempted to create taxonomy of business functions, processes, services, and data? If so, how far along are they?

Back to Top

SERVICE ARCHITECTURE GUIDANCE

Service Architecture starts with the engagement kickoff and educating the organization in the concepts of SOA engineering. It also helps familiarize them with the terminology that will be used throughout the engagement. The workshops covered in Service Architecture include the following:

- a kickoff aimed at introducing the organization to SOA engineering
- Service Architecture workshop to define a common baseline Service Architecture and Governance

Kickoff

Prepare and conduct a kickoff meeting with all attendees of the engagement. During the kickoff, the audience will need to be introduced to the scope of the engagement and the project plan. Ensure everybody agrees to the project plan and is available as needed.

Define Service Architecture

Review the SOA reference architecture of the organization and identify the definition and components of a service, the versioning strategy, and the layered service architecture. Review the enterprise repository meta-model with respect to the lifecycle of a service, all dependent assets and the taxonomies used. Compare the current status with best practices as documented in OUM. Prepare and conduct a workshop on Service Architecture to share the current definition, discuss problem or missing areas and define a common baseline for all further activities. Summarize the findings in a Service Architecture document.

If the organization has already a good maturity on SOA reference architecture or a reference architecture planning engagement has been executed before, this task can be reduced to just presenting the findings as part of the kickoff meeting. However, you should ensure that all participants of the upcoming tasks are familiar with the terminology, taxonomy and basics around services.

Architecture Wrap-Up

Prepare and conduct reviews of the Service Architecture document. Prepare and conduct a short presentation summarizing the changes to be applied. The presentation may be combined with the presentations of the other area wrap-ups into a presentation conducted during the final Service Engineering Wrap-Up.

Key Responsibilities

This area involves the following roles:

Role	Responsibility
Implementing Organization	
Enterprise Architect	Create a presentation for the kickoff and lead the meeting. Review the organization's SOA reference architecture and enterprise repository meta model. Create the presentation for and conduct the Service Architecture workshop. Review the Service Architecture document and the findings presentation. Attend the wrap-up meeting.
Methodologist	Prepare and present an overview of Oracle's enterprise SOA and SOA engineering for the kickoff meeting. Support the preparation of the Service Architecture workshop and create the Service Architecture document. Organize, review and finalize the document. Create and present the finding presentation at the wrap-up meeting.
Customer (User)	
Customer Project Sponsor	Introduce the kickoff workshop sharing the objectives and importance of the project. Attend the wrap-up meeting.
Customer Project Manager	Attend the kickoff meeting and control stakeholder participation.
Enterprise Architect	Prepare and present an overview on the SOA enterprise architecture of the organization for the kickoff meeting. Provide and support the review of the SOA enterprise architecture. Attend the Service Architecture workshop. Review and sign the Service Architecture document. Attend the wrap-up meeting.
Enterprise Repository Manager	Prepare and present an overview on the enterprise repository and the meta model relevant for SOA at the kickoff meeting. Provide and support the review of the enterprise repository. Attend the Service Architecture workshop. Review and sign the Service Architecture document. Attend the wrap-up meeting.
Methodologist	Prepare and present an overview on the SDLC of the organization for the kickoff meeting. Provide and support the review of the method to enforce architectural policies. Attend the Service Architecture workshop. Review the Service Architecture document. Attend the wrap-up meeting.
Configuration Manager	Attend the kickoff meeting and the Service Architecture workshop. Represent the interests of configuration management.
Project Manager	Attend the kickoff meeting and the Service Architecture workshop. Represent the interests of project management.
Business Analyst	Attend the kickoff meeting and the Service Architecture workshop. Represent the interests of business analysis.
System Architect	Attend the kickoff meeting and the Service Architecture workshop. Represent the interests of system architecture.
Designer	Attend the kickoff meeting and the Service Architecture workshop. Represent the interests of software design.
Developer	Attend the kickoff meeting and the Service Architecture workshop. Represent the interests of development.
Quality Manager	Attend the kickoff meeting and the Service Architecture workshop. Represent the interests of quality management.
IS Operations Staff	Attend the kickoff meeting and the Service Architecture workshop. Represent the interests of operations.

Techniques

The following activities are available to assist in this area:

Technique Guidance	Technique Note
Service Architecture	This technique provides assistance on service definition and architectural policy management.
Service Lifecycle	This technique provides assistance on the service lifecycle and the asset types needed for SOA.
IT Asset Management	This technique provides assistance on the meta model for the asset types needed for SOA.

Back to Top

STM.050 - Conduct Team Orientation

Kickoff Workshop (SOA Engineering Overview)

You start the engagement using a kickoff workshop. During this workshop, you should introduce the focus areas of the SOA Engineering Approach and the scope of SOA engineering. Familiarize the participants with the terminology that will be used throughout the engagement.

The workshop will also be used to detail the scope of SOA engineering to be implemented with the engagement, as this may vary from one organization to the next.

INPUT: Overview on Service Engineering.ppt

OUTPUT: Organization has a clear understanding of what is SOA engineering and what is not. Organization has a clear understanding of the high-level SOA Engineering Approach.

Back to Top

EA.010 - Review Architecture Principles, Guidelines and Standards

Initial principles, standards and guidelines for SOA should be determined as part of a SOA Engineering Planning Engagement. Often the first standards are identified during a Service Architecture Workshop where you also look into the SOA Reference Architecture and the Services Implementation Strategy. Please see the task guidance for the EA.075, Develop Technology Reference Architecture, below for more details about such a workshop.

Back to Top

EA.075 Develop Technology Reference Architectures

When determining the Technology Reference Architecture for SOA – called the SOA Reference Architecture – there are certain elements that should be covered. For any situation that involves a set of SOA services of any significant size, or if the enterprise has embraced SOA as an IT strategy, it is important to determine and document a consistent strategy for service implementation. Failure to establish such a strategy might result in an inconsistent service implementation and may negate the advantages of using a SOA approach by greatly hampering reuse.

Therefore, during this task, you define the SOA Reference Architecture that will be used at the enterprise level, one of the most important parts being the definition of the layers of the SOA architecture. The SOA Reference Architecture should describe how services should be implemented in the enterprise architecture, what kind of services should be located where (in what layer or tier), and what access and communication mechanisms should be used as well as define a target for implementation over a few years.

The SOA Reference Architecture should cover:

1. Service Definition - What is a service, service scope, service lifecycle, etc.?
2. Conceptual View - service types and layers (for example, connectivity, data, decision, business, business process, presentation services and associated rules and guidelines for each layer or service type
3. SOA Infrastructure - covers components (for example, a service bus) to perform service routing, service orchestration, governance, management, etc.
4. SOA Security - includes service authorization, identity federation, etc.
5. Implementation - tools and technology for implementing the previous points

6. Deployment - covers service packaging, and a service deployment strategy
7. Patterns and Design Guidelines - includes best practices, and design patterns

Guidelines for points 1 and 2 above are available in the Service Architecture technique.

In a SOA Engineering Planning engagement, the SOA Reference Architecture should be determined during a Service Architecture Workshop, where you also look into Enterprise Architecture Principles, Guidelines and Standards (EA.010) as well as the SOA Reference Architecture.

Service Architecture Workshop

As for every workshop, you need to be well prepared. Ensure that you have someone delivering the workshop with a good knowledge about Service Architecture. Read the Service Architecture technique as a preparation. In the end, the organization needs to decide the course that they will take with respect to Service Architecture.

It is recommended that you start the workshop with a high level overview of the OUM, as well as the recommended approach to SOA engineering. You should base this on the information in the technique guide. Continue to facilitate discussion amongst the enterprise participants to define a SOA Reference Architecture and from there, a detailed set of standards and guidelines with respect to service architecture. You need to be well prepared to participate in the discussion and provide guidance whenever required in addition to facilitating the workshop.

During the workshop preparation you should think forward with respect to an enterprise-specific SOA approach and during the workshop guide the organization against making decisions that may break other areas of the approach that have not been discussed yet. This workshop should be scheduled to last for a full day.

INPUT: Service Architecture technique

OUTPUT: Understanding of the customized approach regarding Service Architecture for which the organization has agreed to adopt.

At this point, note format is appropriate for capturing the output from the workshop.

Document Service Architecture Findings

Formalize the notes taken from the workshop into a document covering the SOA Reference Architecture, and Service Architecture standards, policies and guidelines agreed upon by the organization. You should also formulate the Services Implementation Strategy that describes the strategy on how to move from the current situation to the future Service Architecture. You may use the Service Architecture technique as a template to speed the delivery of this document. Ensure that all aspects of the technique guide that are not relevant to the organization are removed from the document delivered to the organization.

INPUT: Notes from the Service Architecture Workshop

OUTPUT: Service Architecture Document

The SOA Reference Architecture template can be used to document this task.

Back to Top

ER.180.1 Prepare for and Present Solution Recommendation

Reviews

A SOA Engineering Planning engagement is performed through a number of workshops, and where the workshop results are documented into a proper format that will be used for the rest of the engagement. Each workshop result must be presented back to the organization to ensure that it captures their perspective provided in the workshops. The organization should be expected to review the documentation prior to the review presentation and will be given time to provide additional feedback during the review.

INPUT: Documents delivered as result of one or multiple workshops.

OUTPUT: Feedback Presentation

Back to Top

SERVICE ANALYSIS GUIDANCE

During Service Analysis you develop the Service Analysis portion of the enterprise SOA approach. The procedures and policies covered by Service Analysis coincide closely with the types of activities typical of traditional software engineering analysis practices but additionally take into account the needs of the SOA. The workshops covered in Service Analysis are aimed at defining the approach to the following:

- Requirements Management
- Service Identification and Discovery
- SOA Release Planning

Define Approach to Requirements Management

Review the current approach to requirements management and compare with best practices as documented in OUM. Prepare and conduct a workshop on requirements management to share the current requirements analysis process, discuss problem or missing areas and define an approach to be used for SOA engineering. Summarize the findings in a SOA Requirements Process document.

Define Approach to Service Identification and Discovery

Review the current approach to service identification and discovery and compare with best practices as documented in OUM. Prepare and conduct a workshop on service identification and discovery to share the current analysis process, discuss problem or missing areas and define an approach to be used for SOA engineering. Summarize the findings in a Service Identification and Discovery Process document.

Define Approach to SOA Release Planning

Review the current approach to release planning and discovery and compare with best practices as documented in OUM. Prepare and conduct a workshop on SOA release planning to share the current planning process, how dependencies are managed across service reuse and discuss problem or missing areas and define an approach to be used for SOA engineering. Summarize the findings in a SOA Release Planning Process document.

Define Instrumentation Data and Metrics for Service Analysis

When you have defined the approaches to Requirements, Management, Service Identification and Discovery, and SOA Release Planning, you should be able to determine what kind of information can be collected during the different processes and be used as instrumentation data. You would typically do this in separate workshops, where you first determine how you want to instrument the Service Analysis area, and thereafter determine what kind of metrics you need to measure the efficiency of the Service Analysis processes.

Service Analysis Wrap-Up

Prepare and conduct reviews of the documents created. Prepare and conduct a short presentation summarizing the changes to be applied and where instrumentation should be included. The presentation may be combined with the presentations of the other area wrap-ups into a presentation conducted during the final Service Engineering Wrap-Up.

Key Responsibilities

This area involves the following roles:

Role	Responsibility
Implementing Organization	
Enterprise Architect	Support preparations for the workshops. Attend the workshops and review the finding documents. Attend the wrap-up meeting.
Methodologist	Prepare and conduct the workshops. Create the documents and organize the document reviews. Create and present the finding presentation at the wrap-up meeting.
Customer (User)	
Customer Project Sponsor	Attend the wrap-up meeting.
Customer Project Manager	Control stakeholder participation.
Enterprise Architect	Attend the workshops representing the interests of enterprise architecture. Review and sign the documents. Attend the wrap-up meeting.
Enterprise Repository Manager	Attend the workshops representing the interests of asset management. Review and sign the documents. Attend the wrap-up meeting.
Methodologist	Attend the workshops representing the interests of SDLC methodology. Review and sign the documents. Attend the wrap-up meeting.
Configuration Manager	Attend the SOA Release Planning workshop and review the SOA Release Planning Process document. Represent the interests of configuration management.
Project Manager	Attend the SOA Release Planning workshop and review the SOA Release Planning Process document. Represent the interests of project management.
Business Analyst	Attend the Requirements Management and Service Identification and Discovery workshops and review the documents. Represent the interests of business analysis.
System Architect	Attend the Requirements Management and Service Identification and Discovery workshops and review the documents. Represent the interests of system architecture.
IS Operations Staff	Attend the SOA Release Planning workshop and review the SOA Release Planning Process document. Represent the interests of operations.

Techniques

The following activities are available to assist in this area:

Technique Guidance	Technique Note
Service Architecture	This technique provides assistance on service definition and architectural policy management.
Service Lifecycle	This technique provides assistance on the service lifecycle and the asset types needed for SOA.
IT Asset Management	This technique provides assistance on the meta model for the asset types needed for SOA.
Enterprise Requirements Management	This technique provides assistance on requirements management.
Service Identification and Discovery	This technique provides assistance on service identification and discovery.
Service Boundary Analysis	This technique provides assistance on service boundary analysis.
SOA Release Planning	This technique provides assistance on release planning in SOA.
Service Engineering Process Monitoring	This technique provides assistance on how to instrument and what kind of metrics may be relevant for Service Analysis.

Back to Top

EA.095 – Review Enterprise Software Development Process

During this task you should review the organizations software development process, and determine what kind of adjustments would be necessary to support the services development process. Many enterprises struggle with inconsistency across deliveries for projects that need to coexist, which also is an inhibitor to reuse and agility and is addressed by the adoption of a sound enterprise-class SOA approach.

Enterprise SOA Engineering and Management requires that the method for delivering projects that will utilize SOA must be executed with a consistent strategy from inception through delivery and into steady state. Therefore in this task, we will further analyze and improve analysis, delivery and management of services within the organization's software delivery process.

Assess Customer Software Development Process

This step is about gathering the current software development process of the organization. Try to get written information on the current development processes and the guidelines and best practices on software development. Identify how the organization has mapped SOA engineering to his processes and guidelines. The As-Is situation will be further analyzed in the next steps and will be validated against the specific requirements arising from SOA projects.

If an assessment of the organization's SOA strategy was done prior to this engagement, review the results of the assessment with respect to SOA engineering to identify the weaknesses and strengths in the current approach.

Identify Gaps or Areas for Improvement

There are a number of areas of the organization software development process that should be investigated in detail for this type of engagement:

- Service Analysis:
  - Requirements Management (EA.095.1)
  - Service Identification and Discovery (EA.095.2)
  - Service Release Planning (EA.095.3)

For each of the areas where there is a need for improvement, it is recommended that you use a workshop to determine the possible improvements and prepare estimates.

Refine Requirements Management

Review the requirements management process of the organization. Analyze how the organization is managing requirements and models across projects. SOA intends to create reuse across projects across the Enterprise. For that purpose it is important to manage project requirements on Enterprise level. This will enable the projects to discover commonality and reuse on requirements level already.

For mapping business strategy to a service portfolio, it is important to manage the dependencies between requirements, models, services and projects centrally based on a business taxonomy. The Enterprise Requirements Management technique provides leading practices on this topic.

Conduct a workshop on requirements management to create a To-be picture on Requirements Management together with the stakeholders of the organization. Then update the organization's approach according to the findings of the workshop.

INPUT: Enterprise Requirements Management technique

OUTPUT: Understanding of the customized approach regarding SOA Requirements Management, for which the organization has agreed to adopt. At this point, note format is appropriate for capturing the output from the workshop.

Document SOA Requirements Management Findings

Formalize the notes taken from the workshop into a document covering the SOA Requirements Management processes, procedures and guidelines agreed upon by the organization.

INPUT: Notes from the SOA Requirements Management Workshop

OUTPUT: SOA Requirements Management Document

Refine Service Identification and Discovery Approach

Review the analysis process of the organization in Envision and in Implement with respect to Service Identification and Discovery. Analyze the approach chosen and try to gather business alignment created with the approach.

In a successful SOA, creation of new services and reuse goes hand in hand. Experience has shown that it is not valuable to delivery all functionalities as services. On the other hand, some of the functionality that currently resides inside applications should be surfaced as shared services.

It is very important that the analysis process allows, and perhaps even enforces, reuse as early as possible to ensure a good result from the service implementation. To prevent service sprawl, new services should be clearly justified against benefits and risks before being implemented. Please review the Service Identification and Discovery technique on best practices and guidelines.

Conduct a Workshop on Service Identification and Discovery to decide with the organization how to improve their process. Then update the organization's approach according to the findings of the workshop.

INPUT: Service Identification and Discovery technique

OUTPUT: Understanding of the customized approach regarding Service Identification and Discovery, for which the organization has agreed to adopt. At this point, note format is appropriate for capturing the output from the workshop.

Document Service Identification and Discovery Findings

Formalize the notes taken from the workshop into a document covering the Service Identification and Discovery processes, procedures and guidelines agreed upon by the organization.

INPUT: Notes from the Service Identification and Discovery Workshop

OUTPUT: Service Identification and Discovery Document

Refine Service Release Planning Discipline

Review the release planning process of the organization. Analyze how the organization is solving dependencies across projects arising from SOA.

To support business agility it is important to allow for a flexible release management for services. But the more reuse SOA creates, the more dependencies need to be managed. Instead of synchronizing the enterprise on major releases over the year, the release management should allow for parallel deployment of several versions on a service in order to decouple the lifecycles of the applications and systems from the service lifecycles. The SOA Release Planning technique provides more information on this topic.

Conduct a Workshop on SOA Release Planning to decide with the organization how to improve their process. Then update the organization's approach according to the findings of the workshop.

INPUT: SOA Release Planning technique

OUTPUT: Understanding of the customized approach regarding SOA Release Planning, for which the organization has agreed to adopt. At this point, note format is appropriate for capturing the output from the workshop.

Document SOA Release Planning Findings

Formalize the notes taken from the workshop into a document covering the SOA Release Planning processes, procedures and guidelines agreed upon by the organization.

INPUT: Notes from the SOA Release Planning Workshop

OUTPUT: SOA Release Planning Document

Back to Top

BA.017.1 Determine Metrics Collection and Reporting Strategy

When you execute this task at this point of time, you already know the area of focus, which is Service Analysis. Of course, you can detail the area even more if you want to limit the task to cover only a subset of Service Analysis, for example, only Service Identification and Discovery. It all depends on what kind of pain points exists for Service Analysis, and what kind of goals the organization has related to the service engineering process.

The Service Engineering Process Monitoring technique contains guidelines on both what kind of instrumentation data may be relevant for Service Analysis, as well as a set of metrics that may be used as a starting point when executing this task.

Back to Top

ER.180.2 Prepare for and Present Solution Recommendation

Reviews

A SOA Engineering Planning engagement is performed through a number of workshops, and where the workshop results are documented into a proper format that will be used for the rest of the engagement. Each workshop result must be presented back to the organization to ensure that it captures their perspective provided in the workshops. The organization should be expected to review the documentation prior to the review presentation and will be given time to provide additional feedback during the review.

INPUT: Documents delivered as result of one or multiple workshops.

OUTPUT: Feedback Presentation

Back to Top

SERVICE DELIVERY GUIDANCE

Service Delivery covers the core engineering processes and guidelines required to define, design, construct and test services. The workshops covered in Service Delivery are aimed at defining the approach to the following:

- Service Definition
- Service Design
- Service Implementation

- Service Testing

Define Approach to Service Definition

Review the current approach to service definition and the meta model on service contracts and compare with best practices as documented in OUM. Prepare and conduct a workshop on service definition to share the current definition process, the current use of service contracts, discuss problem or missing areas and define an approach to be used for SOA engineering. Summarize the findings in a Service Definition Process document.

Define Approach to Service Design

Review the current approach to service design and the meta model on service interfaces and compare with best practices as documented in OUM. Prepare and conduct a workshop on service design to share the current design process, service design guidelines, discuss problem or missing areas and define an approach to be used for SOA engineering. Summarize the findings in a Service Design Process document.

Define Approach to Service Implementation

Review the current approach to service implementation and the meta model on service implementation and compare with best practices as documented in OUM. Prepare and conduct a workshop on service implementation to share the current implementation process, implementation guidelines and discuss problem or missing areas and define an approach to be used for SOA engineering. Summarize the findings in a Service Implementation Process document.

Define Approach to Service Testing

Review the current approach to service testing and the meta model on test plans and test cases and compare with best practices as documented in OUM. Prepare and conduct a workshop on service testing to share the testing process, discuss service testing strategies and discuss problem or missing areas and define an approach to be used for SOA engineering. Summarize the findings in a Service Testing Process document.

Define Instrumentation Data and Metrics for Service Delivery

When you have defined the approaches to Service Definition, Service Design, Service Implementation and Service Testing, you should be able to determine what kind of information can be collected during the different processes and be used as instrumentation data. You would typically do this in separate workshops, where you first determine how you want to instrument the Service Delivery area, and thereafter determine what kind of metrics you need to measure the efficiency of the Service Delivery processes.

Service Delivery Wrap-Up

Prepare and conduct reviews of the documents created. Prepare and conduct a short presentation summarizing the changes to be applied and where instrumentation should be included. The presentation may be combined with the presentations of the other area wrap-ups into a presentation conducted during the final Service Engineering Wrap-Up.

Key Responsibilities

This area involves the following roles:

Role	Responsibility
Implementing Organization	
Enterprise Architect	Support preparations for the workshops. Attend the workshops and review the finding documents. Attend the wrap-up meeting.
Methodologist	Prepare and conduct the workshops. Create the documents and organize the document reviews. Create and present the finding

	presentation at the wrap-up meeting.
Customer (User)	
Customer Project Sponsor	Attend the wrap-up meeting.
Customer Project Manager	Control stakeholder participation.
Enterprise Architect	Attend the workshops representing the interests of enterprise architecture. Review and sign the documents. Attend the wrap-up meeting.
Enterprise Repository Manager	Attend the workshops representing the interests of asset management. Review and sign the documents. Attend the wrap-up meeting.
Methodologist	Attend the workshops representing the interests of SDLC methodology. Review and sign the documents. Attend the wrap-up meeting.
System Architect	Attend the workshops and review the documents. Represent the interests of system architecture.
Designer	Attend the Service Definition and Service Design workshops and review the documents. Represent the interests of software design.
Developer	Attend the workshops and review the documents. Represent the interests of development.
Quality Manager	Attend the workshops and review the documents. Represent the interests of quality management.

Techniques

The following activities are available to assist in this area:

Technique Guidance	Technique Note
Service Architecture	This technique provides assistance on service definition and architectural policy management.
Service Lifecycle	This technique provides assistance on the service lifecycle and the asset types needed for SOA.
IT Asset Management	This technique provides assistance on the meta model for the asset types needed for SOA.
Enterprise Requirements Management	This technique provides assistance on requirements management.
Service Definition	This technique provides assistance on service definition.
Service Boundary Analysis	This technique provides assistance on service boundary analysis.
Service Design	This technique provides assistance on service design.
Service Implementation	This technique provides assistance on service implementation.
Service Testing	This technique provides assistance on service testing.
Service Engineering Process Monitoring	This technique provides assistance on how to instrument and what kind of metrics may be relevant for Service Delivery.

Back to Top

EA.095 – Review Enterprise Software Development Process

During this task you should review the organization software development process, and determine what kind of adjustments would be necessary to support the services development process. Many enterprises struggle with inconsistency across deliveries for projects that need to coexist, which also is an inhibitor to reuse and agility and is addressed by the adoption of a sound enterprise-class SOA approach.

Enterprise SOA Engineering and Management requires that the method for delivering projects that will utilize SOA must be executed with a consistent strategy from

inception through delivery and into steady state. Therefore in this task we will further analyze and improve analysis, delivery and management of services within the organization's software delivery process.

Assess Customer Software Development Process

This step is about gathering the current software development process of the organization. Try to get written information on the current development processes and the guidelines and best practices on software development. Identify how the organization has mapped SOA engineering to his processes and guidelines. The As-is situation will be further analyzed in the next steps and will be validated against the specific requirements arising from SOA projects.

If an assessment of the organization's SOA strategy was done prior to this engagement, review the results of the assessment with respect to SOA engineering to identify the weaknesses and strengths in the current approach.

Identify Gaps or Areas for Improvement

There are a number of areas of the organization software development process that should be investigated in detail for this type of engagement:

- Service Delivery:
  - Service Definition (EA.095.4)
  - Service Design (EA.095.5)
  - Service Implementation (EA.095.6)
  - Service Testing (EA.095.7)

For each of the areas where there is a need for improvement, it is recommended that you use a workshop to determine the possible improvements and prepare estimates.

Refine Service Definition

Review the solution definition process of the organization regarding service definition. Analyze how the organization identifies the boundaries and granularity of the services. Ensure the organization is familiar with the concept of service contracts. Analyze how the contracts are defined and managed.

Gather how the organization ensures services meet expectations from business. Test the approach the organization has chosen on some sample requirements in parallel with two different teams and check both teams identify similar services with the same boundaries. In order to enable reuse it is important that all the projects spilt requirements into services along similar boundaries.

For more information review the Service Boundary Analysis technique.

Conduct a Workshop on Service Definition to decide with the organization how to improve their process. Then update the organization's approach according to the findings of the workshop.

INPUT: Service Boundary Analysis technique.

OUTPUT: Service Definition which the organization has agreed to adopt. At this point, note format is appropriate for capturing the output from the workshop.

Document Service Definition Findings

Formalize the notes taken from the workshop into a document covering the Service Definition processes, procedures and guidelines agreed upon by the organization.

INPUT: Notes from the Service Definition Workshop

OUTPUT: Service Definition Document

Refine Service Design Process

Analyze the design process of the organization with respect to service design. Analyze how the style of the service interface is gathered and published.

Service interfaces are the meeting point between service consumers and providers. Therefore particular attention should be given to the design of interfaces, especially since it is likely to be called by consumers of different types and technology. Please review the Service Design technique for best practices on service design.

Conduct a Workshop on Service Design to decide with the organization how to improve their process. Then update the organization's approach according to the findings of the workshop.

INPUT: Service Design technique.

OUTPUT: Understanding of the customized approach regarding Service Design, for which the organization has agreed to adopt. At this point, note format is appropriate for capturing the output from the workshop.

Document Service Design Findings

Formalize the notes taken from the workshop into a document covering the Service Design processes, procedures and guidelines agreed upon by the organization.

INPUT: Notes from the Service Design Workshop

OUTPUT: Service Design Document

Refine Service Implementation Process

Analyze the implementation process of the organization. Analyze how services are implemented and how the organization ensures quality of delivery from implementation.

Check if the organization is familiar with unit testing and analyze how this is applied or even enforced in the process. More on this topic can be found in the Service Implementation technique.

Conduct a Workshop on Service Design to decide with the organization how to improve their process. Then update the organization's approach according to the findings of the workshop.

INPUT: Service Implementation technique.

OUTPUT: Understanding of the customized approach regarding Service Implementation, for which the organization has agreed to adopt. At this point, note format is appropriate for capturing the output from the workshop.

Document Service Implementation Findings

Formalize the notes taken from the workshop into a document covering the Service Implementation processes, procedures and guidelines agreed upon by the organization.

INPUT: Notes from the Service Implementation Workshop

OUTPUT: Service Implementation Document

Refine Service Testing Concept

Analyze the testing concept of the organization. Analyze how the organization covers service based testing in collaboration with testing of projects and systems.

Gather the method the organization uses to validate the functionality of a service. Some enterprises spend a lot of time on project integration testing. This can be improved by designing test strategies on service level. Please review the Service Testing technique on the concept of service-based testing.

Conduct a Workshop on Service Testing to decide with the organization how to improve their process. Then update the organization's approach according to the findings

of the workshop.

INPUT: Service Testing technique

OUTPUT: Understanding of the customized approach regarding Service Testing, for which the organization has agreed to adopt. At this point, note format is appropriate for capturing the output from the workshop.

Document Service Testing Findings

Formalize the notes taken from the workshop into a document covering the Service Testing processes, procedures and guidelines agreed upon by the organization.

INPUT: Notes from the Service Testing Workshop

OUTPUT: Service Testing Document

Back to Top

BA.017.2 Determine Metrics Collection and Reporting Strategy

When you execute this task at this point of time, you already know the area of focus, which is Service Delivery. Once again, you can pinpoint the area even more if you want to limit the task to cover only a subset of Service Delivery, for example, only Service Definition. It all depends on what kind of pain points exists for Service Delivery, and what kind of goals the organization has related to the service engineering process.

The Service Engineering Process Monitoring technique contains guidelines on both what kind of instrumentation data may be relevant for Service Delivery, as well as a set of metrics that may be used as a starting point when executing this task.

Back to Top

ER.180.3 Prepare for and Present Solution Recommendation

Reviews

A SOA Engineering Planning engagement is performed through a number of workshops, and where the workshop results are documented into a proper format that will be used for the rest of the engagement. Each workshop result must be presented back to the organization to ensure that it captures their perspective provided in the workshops. The organization should be expected to review the documentation prior to the review presentation and will be given time to provide additional feedback during the review.

INPUT: Documents delivered as result of one or multiple workshops.

OUTPUT: Feedback Presentation

Back to Top

SERVICE MANAGEMENT GUIDANCE

Service Management covers the activities to transition and operate services. The workshops covered in Service Management are aimed at defining the approach to the following:

- Service Deployment
- Service Operations, Administration and Maintenance (OA&M)

Define Approach to Service Deployment

Review the current approach to service deployment and the deployment technique and compare with best practices as documented in OUM. Prepare and conduct a workshop on service deployment to share the current deployment strategy and discuss problem or missing areas and define an approach to be used for SOA engineering. Summarize the findings in a Service Deployment Strategy document.

Define Approach to Service Operations, Administration and Maintenance (OA&M)

Review the current approach to service OA&M including policies and procedures and compare with best practices as documented in OUM. Prepare and conduct a workshop on service OA&M to share policies and procedures and discuss problem or missing areas and define an approach to be used for SOA engineering. Summarize the findings in a Service OA&M Policies and Procedures document.

Define Instrumentation Data and Metrics for Service Delivery

When you have defined the approaches to Service Deployment and Service Operations, Administration and Maintenance (OA&M), you should be able to determine what kind of information can be collected during the different processes and be used as instrumentation data. You would typically do this in separate workshops, where you first determine how you want to instrument the Service Management area, and thereafter determine what kind of metrics you need to measure the efficiency of the Service Management processes.

Service Management Wrap-Up

Prepare and conduct reviews of the documents created. Prepare and conduct a short presentation summarizing the changes to be applied and where instrumentation should be included. The presentation may be combined with the presentations of the other area wrap-ups into a presentation conducted during the final Service Engineering Wrap-Up.

Key Responsibilities

This area involves the following roles:

Role	Responsibility
Implementing Organization	
Enterprise Architect	Support preparations for the workshops. Attend the workshops and review the finding documents. Attend the wrap-up meeting.
Methodologist	Prepare and conduct the workshops. Create the documents and organize the document reviews. Create and present the finding presentation at the wrap-up meeting.
Customer (User)	
Customer Project Sponsor	Attend the wrap-up meeting.
Customer Project Manager	Control stakeholder participation.
Enterprise Architect	Attend the workshops representing the interests of enterprise architecture. Review and sign the documents. Attend the wrap-up meeting.
Enterprise Repository Manager	Attend the workshops representing the interests of asset management. Review and sign the documents. Attend the wrap-up meeting.

Methodologist	Attend the workshops representing the interests of SDLC methodology. Review and sign the documents. Attend the wrap-up meeting.
System Architect	Attend the Service Deployment workshop and review the Service Deployment Strategy document. Represent the interests of system architecture.
Designer	Attend the workshops and review the documents. Represent the interests of software design.
Developer	Attend the Service Deployment workshop and review the Service Deployment Strategy document. Represent the interests of development.
Quality Manager	Attend the Service Deployment workshop and review the Service Deployment Strategy document. Represent the interests of quality management.
IS Operations Staff	Attend the workshops and review the documents. Represent the interests of operations.

Techniques

The following activities are available to assist in this area:

Technique Guidance	Technique Note
Service Architecture	This technique provides assistance on service definition and architectural policy management.
Service Lifecycle	This technique provides assistance on the service lifecycle and the asset types needed for SOA.
IT Asset Management	This technique provides assistance on the meta model for the asset types needed for SOA.
Service Packaging and Deployment	This technique provides assistance on service packaging and deployment.
Service Engineering Process Monitoring	This technique provides assistance on how to instrument and what kind of metrics may be relevant for Service Management.

Back to Top

EA.095 – Review Enterprise Software Development Process

During this task you should review the organization software development process, and determine what kind of adjustments would be necessary to support the services development process. Many enterprises struggle with inconsistency across deliveries for projects that need to coexist, which also is an inhibitor to reuse and agility and is addressed by the adoption of a sound enterprise-class SOA engineering.

Enterprise SOA Engineering and Management requires that the method for delivering projects that will utilize SOA must be executed with a consistent strategy from inception through delivery and into steady state. Therefore in this task we will further analyze and improve analysis, delivery and management of services within the organization's software delivery process.

Assess Customer Software Development Process

This step is about gathering the current Software Development Process of the organization. Try to get written information on the current development processes and the guidelines and best practices on software development. Identify how the organization has mapped SOA engineering to his processes and guidelines. The As-is situation will be further analyzed in the next steps and will be validated against the specific requirements arising from SOA projects.

If an assessment of the organization's SOA strategy was done prior to this engagement, review the results of the assessment with respect to SOA engineering to identify the weaknesses and strengths in the current approach.

Identify Gaps or Areas for Improvement

There are a number of areas of the organization software development process that should be investigated in detail for this type of engagement:

- Service Management
  - Service Packaging and Deployment (EA.095.8)
  - Service Operations, Administration and Management (OA&M) (EA.095.9)

For each of the areas where there is a need for improvement, it is recommended that you use a workshop to determine the possible improvements and prepare estimates.

Refine Service Packaging and Deployment Strategy

In this step analyze the deployment strategy of the organization. Check the strategy for services in the different staging environments and analyze, if the packaging strategy support for decoupled service lifecycles.

In order to enable business agility, services should be managed as entities on its own. Consider a service to be a smaller sized application that can be packaged and deployed as independent unit, which is very important for decoupling the service lifecycle from the lifecycle of applications. The Service Packaging and Deployment technique provides more information on this.

Conduct a Workshop on Service Testing to decide with the organization how to improve their process. Then update the organization's approach according to the findings of the workshop.

INPUT: Service Packaging and Deployment technique

OUTPUT: Understanding of the customized approach regarding Service Deployment Strategy for which the organization has agreed to adopt. At this point, note format is appropriate for capturing the output from the workshop.

Document Service Architecture Findings

Formalize the notes taken from the workshop into a document covering the Service Deployment Strategy, processes, standards, policies and guidelines agreed upon by the organization.

INPUT: Notes from the Service Deployment Strategy Workshop

OUTPUT: Service Deployment Strategy Document

Refine Operations, Administration and Management (OA&M) Strategy

Analyze the OA & M strategy of the organization. Gather how the organization manages services and depending assets at design and runtime.

In order to enable run-time governance, the operations team must be able to monitor Service Level Agreements (SLA) for services. The SLAs that a service offers to its consumers should be managed as part of a contract. But as services can be shared across several consumers the load created by each consumer also needs to be considered. To do that, the operations team can use the concept of a usage agreement where each consumer will agree upon the terms and conditions of service usage. Monitoring the SLAs for a service therefore includes monitoring of the service provider as well as monitoring of service usage per consumer.

In order to ensure effective collaboration between the various stakeholders in the engineering process, it is recommended to use an Enterprise Repository to manage all the assets (such as services, contracts, requirements, etc.) and their relationships. List the types of assets used during the engineering process as identified in the previous steps and analyze how the assets are managed. Define the asset meta-data needed for collaboration between the stakeholders and if not already available try to derive a meta-model for the SOA engineering assets. Asset meta-data includes asset attributes, classification and relationships. The results should be considered in the context of a Service Portfolio Management initiative which will be responsible for creating, enhancing and maintaining the SOA engineering assets in the Enterprise repository.

Conduct a Workshop on Service Management to decide with the organization how to improve their process. Then update the organization's approach according to the findings of the workshop. You might consider splitting the workshop between design time governance and run-time governance.

OUTPUT: Understanding of the customized approach regarding Service Operations, Administration and Maintenance, for which the organization has agreed to adopt.

Document Service OA&M Findings

INPUT: Notes from the Service OA&M Workshop

OUTPUT: Service OA&M Document

Formalize the notes taken from the workshop into a document covering the Service OA&M processes, procedures and guidelines agreed upon by the organization.

Back to Top

BA.017.3 Determine Metrics Collection and Reporting Strategy

When you execute this task at this point of time, you already know the area of focus, which is Service Management. Once again, you can pinpoint the area even more if you want to limit the task to cover only a subset of Service Management, for example, only Service Deployment. It all depends on what kind of pain points exists for Service Management, and what kind of goals the organization has related to the service engineering process.

The Service Engineering Process Monitoring technique contains guidelines on both what kind of instrumentation data may be relevant for Service Management, as well as a set of metrics that may be used as a starting point when executing this task.

Back to Top

ER.180.4 Prepare for and Present Solution Recommendation

Reviews

A SOA Engineering Planning engagement is performed through a number of workshops, and where the workshop results are documented into a proper format that will be used for the rest of the engagement. Each workshop result must be presented back to the organization to ensure that it captures their perspective provided in the workshops. The organization should be expected to review the documentation prior to the review presentation and will be given time to provide additional feedback during the review.

INPUT: Documents delivered as result of one or multiple workshops.

OUTPUT: Feedback Presentation

Back to Top

WRAP-UP GUIDANCE

During Wrap-Up, all work products are finalized for the engagement and prepared for project sign-off. The completed approach is presented back to the organization to wrap-up the engagement.

Service Engineering Wrap-Up

Review all documents created and ensure consistency across the whole approach. You may want to combine all documents regarding changes to the engineering process into one SOA Engineering Approach document. Similarly, you may want to combine all documents regarding instrumenting each area into a single SOA Engineering Instrumentation and Metrics document.

For the process changes, create a summary presentation providing an overview on the SOA Engineering Approach created and a summary on changes the organization will need to implement. For the instrumentation of the processes, create a summary presentation showing what areas should be monitored with what purpose (what goals and objectives should be measured), and what kind of instrumentation should be used.

Prepare and conduct a wrap-up meeting with all attendees of the engagement. During the wrap-up meeting, collect feedback from the organization and share ideas about next steps.

Key Responsibilities

This area involves the following roles:

Role	Responsibility
Implementing Organization	
Enterprise Architect	Review all documents and support preparations of the summary presentation. Attend the wrap-up meeting.
Methodologist	Review all documents. Create and present the summary presentation at the wrap-up meeting.
Customer (User)	
Cusotmer Project Sponsor	Attend the wrap-up meeting and sign all documents created.
Customer Project Manager	Attend the wrap-up meeting and control stakeholder participation.
Enterprise Architect	Attend the wrap-up meeting.
Enterprise Repository Manager	Attend the wrap-up meeting.
Methodologist	Attend the wrap-up meeting.
System Architect	Attend the wrap-up meeting.
Configuration Manager	Attend the wrap-up meeting.
Project Manager	Attend the wrap-up meeting.
Business Analyst	Attend the wrap-up meeting.
Designer	Attend the wrap-up meeting.
Developer	Attend the wrap-up meeting.
Quality Manager	Attend the wrap-up meeting.
IS Operations Staff	Attend the wrap-up meeting.

Back to Top

EA.095.10 – Review Enterprise Software Development Process

Review and Document Findings

Review the SOA engineering method defined with the organization. Check the changes agreed upon have been documented and the organization has assigned responsibilities for implementing the changes.

Consider creating a document that summarizes the software development process for services as defined in this task or at least the changes to the current development process. Forward this document to all stakeholders for review to ensure consistency in the new approach.

Back to Top

BA.017.4 Determine Metrics Collection and Reporting Strategy

When you execute this task at this point of time, it is about wrapping up what you have already determined as part of the previous areas. Execute the task step, Determine metrics collection and reporting strategy. Review the result from this step for each area. It is recommended that you combine all the results into a single document, and create a overview in the front summarizing the various means for tracking progress toward the identified goals, including what kind of metrics will be required, and how the result should be reported. This is a necessary step toward verifying consistency between the processes and metrics notations.

Back to Top

ER.180.5 Prepare for and Present Solution Recommendation

Reviews

A SOA Engineering Planning engagement is performed through a number of workshops, and where the workshop results are documented into a proper format that will be used for the rest of the engagement. Each workshop result is presented back to the organization to ensure that it captures their perspective provided in the workshops. The organization should be expected to review the documentation prior to the review presentation and will be given time to provide additional feedback during the review.

INPUT: Documents delivered as result of one or multiple workshops.

OUTPUT: Feedback Presentation

The discussion following this end presentation of the engagement may identify some questions or challenges that need to be addressed for the engagement to be accepted. It should be positioned, during the presentation, that the enterprise executive sponsor takes responsibility for the resolution of any such unresolved items. This alleviates any need to get the enterprise executive audience back together to deliver another version of the presentation, and expedites the completion of deliverables by creating a single interface point. When the take-away items are completed to the satisfaction of the executive sponsor, the engagement project leader should obtain written approval of the acceptance of the engagement. This acceptance is done as part of Post Delivery, task SM.050 - Gain Acceptance.

Back to Top

POST DELIVERY GUIDANCE

Following the successful completion of an engagement, a few post-delivery activities should take place, i.e., acceptance sign-off and documenting lessons learned.

CMM.040 Document Lessons Learned

After the project has been delivered and accepted any feedback should be summarized and submitted to Oracle's Global Methods team at ominfo_us@oracle.com. Feedback should include the following:

- overall experience, both positive and negative regarding the use of the Planning view
- major challenges and roadblocks encountered during service delivery and how these issues were addressed
- suggested improvements to the delivery process
- suggested improvements to the templates

Back to Top

Copyright © 2006, 2016, Oracle and/or its affiliates. All rights reserved.

SERVICE-ORIENTED ARCHITECTURE (SOA) ENGINEERING PLANNING TECHNIQUE/TOOL GUIDELINES

Back to View

This section describes specific techniques and tools that should be used during Service Engineering Planning. For additional information about techniques/tools, please access the OUM Supplemental Guidance page from the Method Navigation pull down.

ID	Task	Work Product	Technique
Mobilization			
ER.040	Research Enterprise	Enterprise Research Findings	Stakeholder Capturing
ER.020	Determine Envision Engagement Strategy	Envision Engagement Strategy	Service Architecture Motivational Modeling Business Scope Definition Functional or Process Modeling Enterprise Domain Model - Data Ownership Approach Enterprise Domain Model - Data Relationship Approach Service Modeling
ER.020	Determine Envision Engagement Strategy	Envision Engagement Strategy	Service Architecture Motivational Modeling Business Scope Definition Functional or Process Modeling Enterprise Domain Model - Data Ownership Approach Enterprise Domain Model - Data Relationship Approach Service Modeling
ER.030	Set and Agree on Plan for Envision Engagement	Envision Engagement Plan	Service Architecture

			Motivational Modeling Business Scope Definition Functional or Process Modeling Enterprise Domain Model - Data Ownership Approach Enterprise Domain Model - Data Relationship Approach Service Modeling
Service Architecture			**Service Architecture** **Service Lifecycle** **IT Asset Management**
STM.050	Conduct Team Orientation	Project Kickoff Presentation	Service Architecture
EA.010	Review Architecture Principles, Guidelines and Standards	Architecture Principles, Guidelines and Standards	Service Architecture Service Boundary Analysis Service Identification and Discovery
EA.075	Develop Technology Reference Architectures	Technology (SOA) Reference Architectures	Service Lifecycle Service Architecture SOA Release Planning
Service Analysis			**Service Architecture** **Service Lifecycle** **IT Asset Management** **Enterprise Requirements Management** **Service Identification and Discovery** **Service Boundary Analysis** **SOA Release Planning**
EA.095 (.1,.2,.3)	Review Enterprise Software Development Process	Enterprise Software Development Process	Enterprise Requirements Management Service Identification and Discovery SOA Release Planning
BA.017.1	Determine Metrics Collection and Reporting Strategy	Metrics Collection and Reporting Strategy	Service Engineering Processing Monitoring
Service Delivery			**Service Architecture** **Service Lifecycle** **IT Asset Management** **Enterprise Requirements Management** **Service Definition** **Service Boundary Analysis** **SOA Release Planning** **Service Design** **Service Implementation** **Service Testing**
EA.095 (.4,.5,.6,.7)	Review Enterprise Software Development Process	Enterprise Software Development Process	Service Boundary Analysis Service Design Service Implementation Service Testing
BA.017.2	Determine Metrics Collection and Reporting	Metrics Collection and Reporting Strategy	Service Engineering Processing Monitoring

	Strategy		
Service Management			**Service Architecture** **Service Lifecycle** **IT Asset Management** **Service Packaging and Deployment**
EA.095 (.8, .9)	Review Enterprise Software Development Process	Enterprise Software Development Process	Service Packaging and Deployment
BA.017.3	Determine Metrics Collection and Reporting Strategy	Metrics Collection and Reporting Strategy	Service Engineering Processing Monitoring
Wrap-Up			
EA.095.10	Review Enterprise Software Development Process	Enterprise Software Development Process	Enterprise Requirements Management Service Identification and Discovery SOA Release Planning Service Boundary Analysis Service Design Service Implementation Service Testing Service Packaging and Deployment

Back to Top

ORACLE

Copyright © 2006, 2016, Oracle and/or its affiliates. All rights reserved.